Solution Requirements

Date	13 July 2026
Team ID	• AgroSense AI – Intelligent Crop Recommendation System
Project Name	
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	Users can securely access the Crop Recommendation System through the web application.	Medium
2	Authorization levels	General users (farmers) can enter crop parameters and receive recommendations. No separate admin role is implemented.	Low
3	External interfaces	*The system interacts with the trained Machine Learning model and crop dataset to generate recommendations through the Flask web interface.*	High

4	Transactions processing	*The system receives soil parameters (N, P, K, Temperature, Humidity, pH, Rainfall), validates them, sends them to the trained ML model, and generates a crop recommendation in real time.*	*High*
5	Reporting	*Display the recommended crop to the user. The system presents prediction results through the web interface.*	*Medium*
6	Business rules, etc.	*All seven input parameters must be provided within valid ranges. The ML model predicts only from the crops included in the trained dataset. Invalid or missing inputs are rejected.*	*High*
7	Compliance to laws or regulations	*The application does not collect or store personal user information. It is intended for educational and research purposes and follows responsible handling of agricultural data.*	*Low*
8	*Other*	*The application provides a simple, responsive, and user-friendly interface to help farmers quickly obtain crop recommendations.*	*Medium*

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**

1	Performance & Speed	*The system should process user inputs and generate a crop recommendation quickly using the trained machine learning model.*	*Prediction generated within 2–3 seconds.*
2	Scalability	*The application should support multiple users accessing the web interface simultaneously without significant performance degradation.*	*Supports at least 100 concurrent users.*
3	Security & Data Privacy	*[Detail data protection, encryption, and vulnerability management]*	*(e.g., End-to-end AES-256 encryption)*
4	Reliability & Availability	*The application does not store personal user information. User input is processed only for prediction, and server-side validation is performed.*	*No persistent storage of personal data; input validation for all fields.*
5	Usability & Accessibility	*The web interface should be simple, responsive, and easy to use on desktop and mobile devices for farmers and general users.*	*Responsive UI with clear input forms and readable prediction results.*
6	*Other*	*The application should have modular Flask code and a reusable ML prediction module so future crop datasets or models can be integrated easily.*	*Modular code structure that supports future model updates with minimal code changes.*